

Project Briefing: Imputation-Free Multimodal Transformer for Wearable Health Monitoring

1. Project Overview & Clinical Context

Wearable health monitoring devices (like smartwatches and chest straps) fuse multiple physiological signals—such as Electrocardiograms (ECG), Photoplethysmograms (PPG), Accelerometers (ACC), and Electrodermal Activity (EDA)—to monitor patient stress levels and track physical activity in real-time.

However, in everyday settings, sensors frequently fail or drop out due to: * **Motion:** High physical movement shifts the smartwatch, causing optical sensors (PPG) to lose skin contact. * **Sweat:** Sweat blocks skin conductance (EDA) electrodes. * **Battery Management:** The wearable operating system shuts down power-hungry optical sensors to save battery.

When a sensor drops out, standard deep learning models (like CNNs or LSTMs) crash because they expect a fixed-grid input.

2. What Existing Works Do (And Where They Fail)

To keep models from crashing under sensor dropout, existing systems rely on **imputation** (filling in the blanks): * **Zero-Padding:** Filling the missing sensor's slot with flat zeros. This is dangerous; in clinical settings, a flatline ECG represents cardiac arrest, not "missing data." This artificial drop confuses the network. * **Generative AI Imputation:** Training a secondary AI (like a GAN) to "predict" and draw the missing wave. This is computationally slow for low-power smartwatches and runs the risk of "hallucinating" false physiological readings (e.g., drawing normal heart waves when the patient is actually experiencing an emergency). * **Methodological Flaw (Subject Leakage):** Many recent publications report 99% accuracy on datasets like WESAD. However, they achieve this by splitting the data randomly (combining all subjects' data and taking a random 85:15 split). Because physiological baselines are unique to individuals, this allows the model to memorize subject identities. The model fails completely when tested on a new, unseen patient.

3. Our Proposed Solution & Core Novelty

This project introduces a **Hybrid CNN-Transformer** model designed specifically for **dynamic sensor dropout** without data imputation or resampling.

Here are the three novel mechanisms we are implementing:

Novelty 1: Imputation-Free Token Eviction

Instead of filling missing slots with zeros or guessing fake data, our model **completely deletes (evicts)** the missing sensor's token from the sequence. * Because the Transformer's self-attention mechanism processes inputs as an unordered set, the sequence length shrinks dynamically. * When a sensor drops, the computational graph automatically shrinks. This results in **quadratic power savings** on the device without sacrificing accuracy or introducing synthetic noise.

Novelty 2: Heterogeneous multi-rate Tokenization (No Resampling)

Sensors operate at different speeds (e.g., ECG at 700 Hz, EDA at 4 Hz). Traditional approaches force-resample all signals to a single frequency, which either destroys cardiac details or wastes memory. * We use dedicated, individual 1D CNN tokenizers. * These tokenizers use custom strides (step sizes) to compress raw waveforms of different lengths into uniform, high-level features of the exact same size. This allows the model to ingest multi-rate signals natively without uniform resampling.

Novelty 3: Strict Subject-Independent Validation (LOSO)

To ensure our model is clinically viable, we reject random data splits. We evaluate our framework using **Leave-One-Subject-Out (LOSO) Cross-Validation**: * We train the model on $S-1$ subjects and test it on the remaining held-out subject. * We repeat this process for every subject in the dataset. * This ensures that our model learns general, subject-independent physiological dynamics rather than memorizing individual heart rate baselines.

4. Research Questions

1. How can a set-based Transformer architecture be optimized to process variable-length sensor sequences without the need for statistical or generative data imputation under modality dropout?
2. How does using strided 1D convolutions on original sensor sampling rates affect classification accuracy compared to standard signal resampling methods?

3. How does complete token eviction of missing sensors impact the model's classification accuracy, algorithmic complexity, and execution latency during simulation compared to standard data imputation and placeholder-masking methods?

5. Potential Application Areas

- **Clinical Remote Patient Telemonitoring:** Continuous cardiac and autonomic monitoring for home-bound or elderly patients, where electrode shifts or displacements are frequent.
 - **High-Activity Fitness and Athletics:** Stress and physical exertion tracking during sports, where sweat and vigorous motion corrupt individual optical sensors.
 - **Emergency Responder and Military Vitals:** Health and cognitive state monitoring in harsh field environments where sensor damage or battery constraints force selective sensor deactivation.
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6. Summary of Project Deliverables

At the end of this project, we will deliver a simulated Python/PyTorch framework featuring: 1. A synchronized, multi-rate pre-processing pipeline for the benchmark **WESAD** and **PAMAP2** datasets. 2. The Hybrid CNN-Transformer architecture supporting dynamic token eviction. 3. A robustness benchmark evaluating classification performance under simulated sensor dropout rates ranging from 0% to 75%. 4. Inference speed and computational complexity scaling graphs. 5. Attention weight heatmaps showing how the model dynamically shifts its focus away from noisy/missing channels to clean sensors.